PS 26132 — Public Dataset Landscape

Every source, what it actually contains, how to pull it, and exactly where the public data runs out

Market linkages & price discovery for farmers — Govt. of Maharashtra (MSInS)

Prepared for GlobeX — 2 September 2026

●	❍	×
prices & arrivals genuinely public	quality standards published, not dataset-ified	FPO/buyer & storage data does not exist publicly

1 What this problem statement actually needs data for

26132 asks for six things, and public-data coverage is wildly uneven across them. This report goes source-by-source only for what's real — no dataset is listed here without a link and a description of what's actually in it.

	Data need	**Public GoI coverage**	**Verdict**
1	Mandi prices (min/modal/max)	● Strong	Daily, national, years of history, documented API.
2	Arrival volumes	● Strong	Same source, same access path as prices.
3	Quality/grading standards	❍ Partial	The AGMARK grade specifications themselves are public; there is no dataset of actual graded lots.
4	FPO directory & buyer credentials	× Absent	No structured, complete public dataset exists anywhere in government data.
5	Storage/warehouse options	× Mostly absent	WDRA publishes a registered-warehouse list; no public capacity or pricing data.
6	Weather / yield signals (optional enrichment)	❍ Partial	IMD and Bhuvan/ISRO data exist but need separate integration; not core to the MVP.

2 Primary Government of India sources — prices & arrivals

Agmarknet, via the Open Government Data (OGD) platform

Agmarknet itself (`agmarknet.gov.in`) is the Directorate of Marketing & Inspection's public dashboard for mandi prices, but the portal has no clean bulk API of its own — integrations go through its official mirror on `data.gov.in`. Two resources matter here:

- **"Current daily price of various commodities from various markets (Mandi)"** — state, district and market-level daily records: commodity, variety, min/modal/max price (Rs./quintal) and arrival date. This is the workhorse dataset for the price-trend and sale-window features.
- **"Variety-wise Daily Market Prices Data of Commodity"** — the same data broken out at variety granularity (e.g. not just "Wheat" but the specific variety traded), useful when the grading layer needs to match a specific variety's price rather than a commodity average.

```
GET https://api.data.gov.in/resource/9ef84268-d588-465a-a308-a864a43d0070
?api-key={your_key}&format=json&offset=0&limit=100
&filters[state]=Maharashtra&filters[commodity]=Onion
```

Coverage: national, state-wise and market-wise; several years of daily history. **Update frequency:** daily. **Access:** free API key from a `data.gov.in` account (register with a Google or government email); a shared public demo key exists for prototyping but carries tighter rate limits and should not be relied on for a live demo. **Format:** JSON or CSV via the `format` parameter, though some older resources ignore the request and always return CSV — check the response `Content-Type` rather than assuming. **License:** Government Open Data License – India.

AIKosh — the IndiaAI Mission's dataset repository

`aikosh.indiaai.gov.in` is a newer GoI platform, run under the IndiaAI Mission (MeitY), whose explicit purpose is hosting datasets curated for AI training and use — as distinct from data.gov.in's general open-data mandate. It carries a packaged version of the variety-wise mandi price dataset. Citing AIKosh alongside data.gov.in is worth doing deliberately in the pitch: it is the government's own "data curated for AI" platform, which answers a judge's "is this real AI on real data" question about as directly as a citation can.

Don't over-rely on any single mirror

All three of Agmarknet, its data.gov.in mirror, and its AIKosh repackaging trace back to the same underlying DMI/Ministry of Agriculture source. They are not independent datasets — treat them as one source with three access points, and validate whichever one you build the ingestion pipeline against.

3 Non-official mirrors — faster to prototype against

Source	What it actually offers	Best use
CEDA Agri Market Data (Ashoka University CEDA) — `agmarknet.ceda.ashoka.edu.in`	Repackages the same DMI source with cleaner state/district-level exports, Census-2011-mapped boundaries, and built-in time-series/heatmap visualisation and comparison tools.	Fast exploratory analysis and sanity-checking before writing the live ingestion pipeline.
India Data Portal — APMC arrivals and prices (Agmarknet) dataset	A cleaned, versioned CKAN-hosted copy of the arrivals-and-prices data with a documented resource page.	A stable fallback if the live data.gov.in API misbehaves during development or demo prep.
Kaggle mirrors (e.g. "Indian Agricultural Mandi Prices", "Daily Wholesale Commodity Prices – India Mandis")	Community-maintained CSV snapshots of the same underlying data, already flattened.	Offline development and model prototyping before the live API integration is wired up.

4 eNAM — what it is, and what it is not

eNAM is a dashboard, not a data source

`enam.gov.in` is the National Agriculture Market trading platform, and its price dashboard (`enam.gov.in/web/dashboard/agmarknet`) simply re-displays the same Agmarknet/DMI data described above — it is not an independent feed. eNAM does not expose transaction-level trading data or an open API to third parties. Any pitch claiming "eNAM integration" as a built feature will not survive a judge asking to see the integration; the honest framing is a future partnership/onboarding target, not a current data source.

5 The real gaps — where public data simply does not exist

FPO and verified-buyer directories

There is no government dataset — structured or otherwise — listing farmer producer organisations with verifiable attributes (registration status, commodity focus, aggregation capacity) or buyer credentials (procurement history, payment reliability) in a form usable for matching. NABARD publishes some FPO programme information and individual state cooperation departments maintain scattered registers, but none of it is complete, structured, or centrally accessible. **This has to be self-built**: a small, clearly-labelled pilot seed dataset (real-structure, synthetic-content) is the honest path, not a claim of sourced data.

Quality grading

AGMARK grade specifications (the standards themselves — moisture content, size, foreign matter thresholds, etc. per commodity) are published and public. What does not exist is a dataset of *actual graded lots* — no public record of "this specific batch was graded X." Document/OCR verification against the published standard is honest engineering; claiming a trained grading model would require a labelled dataset that has to be built from scratch, most plausibly by photographing and grading a small sample set during prototyping.

Storage and logistics

The Warehousing Development & Regulatory Authority (WDRA) publishes a list of registered warehouses, which gives location and registration status, but not live capacity, current occupancy, or storage pricing. Transport/freight cost data is likewise not published in a usable open form. The existing GlobeX ETA/landed-cost estimator can supply distance and time; storage-cost and live-capacity inputs will need to be assumption-based or manually sourced for the demo, not pulled from a live feed.

6 Data quality risks — and the cleaning checklist that avoids them

Known issue	Why it happens	Cleaning step
Inconsistent commodity/variety naming across states	Each state market board enters its own free-text labels into the same national schema	Build a canonical commodity/variety lookup table; fuzzy-match incoming labels against it before ingestion.
Missing days / gaps in the series	Not every market reports every day, and reporting reliability varies by state	Define an explicit gap-fill policy (carry-forward with a flag, or interpolate) rather than letting gaps silently become zeros or nulls the forecaster misreads.
Non-standard units	Prices are per quintal in most records, but not guaranteed consistent across every state/market entry	Normalise to one unit at ingestion and validate the conversion, don't assume.
API pagination edge case	The API can return an empty `records` array before the reported `total` count is reached	Stop pagination on the first empty response rather than trusting the total-count arithmetic.
Format parameter sometimes ignored	Some older data.gov.in resources always return CSV regardless of the requested `format`	Check the response `Content-Type` header and branch the parser accordingly, rather than assuming JSON.
Field-name case sensitivity in filters	`filters[State]` works, `filters[state]` silently returns nothing on some resources	Confirm exact field casing per resource before relying on server-side filtering; don't assume a naming convention holds across datasets.

Budget real time for this, not model time

None of the issues above are modeling problems. They are exactly the unglamorous data-engineering work that decides whether a live demo works in front of judges. Treat the ingestion/cleaning layer as its own work item with its own time budget, not an afterthought bolted onto the forecasting task.

7 Recommended field mapping into GlobeX's existing schema

The ingestion pipeline's job is to make mandi records look like the trade-panel records the forecaster and ranking engine already expect — same shape, different domain.

Agmarknet/OGD field	Maps to (GlobeX)	Notes
State, District, Market	Corridor origin (region/market level)	A "corridor" becomes a market; no schema change needed, only relabeling.
Commodity, Variety	Product/commodity identifier	Requires the canonical lookup table described above before it can join cleanly to other tables.
Arrival Date	Transaction date / time-series index	Same role it plays for trade data — the axis the forecaster runs against.
Min/Modal/Max Price	Price features (point + spread)	The min–max spread is a genuinely useful extra feature the trade-corridor data didn't have — it's a built-in volatility signal, worth keeping rather than collapsing to modal price alone.
Arrival Quantity	Volume feature	Feeds both the forecaster (demand proxy) and the ranking engine (glut-risk dimension).

Bottom line

The one dataset this problem statement structurally cannot exist without — mandi prices and arrivals — is genuinely, solidly public: documented, API-accessible, updated daily, with GoI-run mirrors (AIKosh) as well as faster third-party ones (CEDA, India Data Portal) for prototyping. Everything else the PS asks for — verified buyers, FPO directories, quality-graded lots, storage capacity — has no public dataset behind it at all, and needs to be honestly scoped as pilot/seed data or a stated roadmap item rather than implied as already sourced. That split is exactly what should drive the build order: ingestion and forecasting first, because the data to do them well actually exists; matching and grading second, built openly on small, labelled seed data rather than quietly on synthetic data dressed up as real.